•[3]:

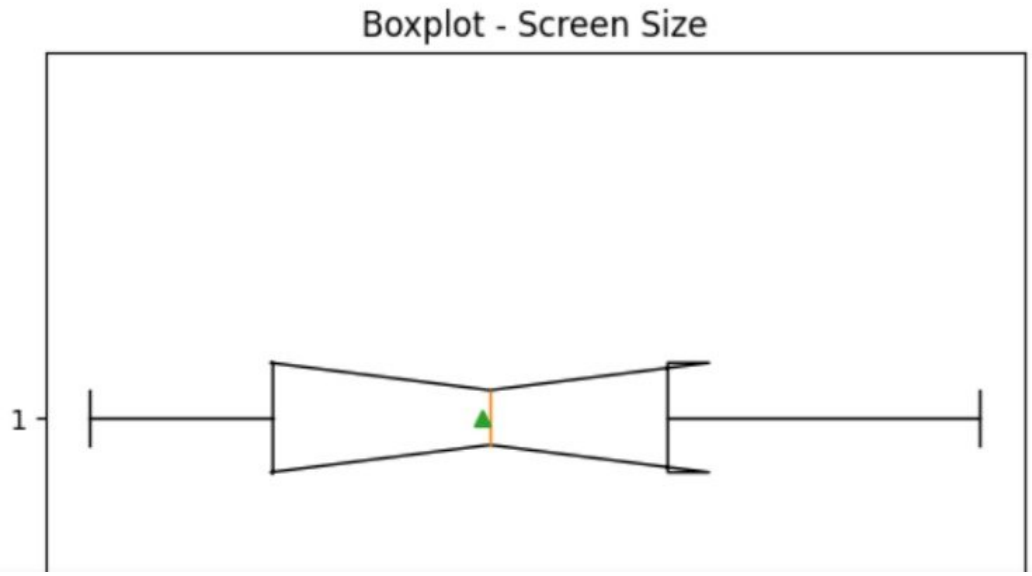
```python
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("laptop.csv")

print("Dataset:")
print(df)
print("\nBoxplot for Screen Size")

plt.boxplot(df["Screen"], notch=True, vert=False, showmeans=True)
plt.xlabel("Screen")
plt.title("Boxplot - Screen Size")
plt.show()
print("\nBoxplot for Weight")

plt.boxplot(df["Weight"])
plt.ylabel("Weight")
plt.title("Boxplot - Weight")
plt.show()


print("\nSeaborn Boxplot for Weight")

sns.boxplot(x=df["Weight"])
plt.title("Seaborn Boxplot - Weight")
plt.show()
```

•[3]:

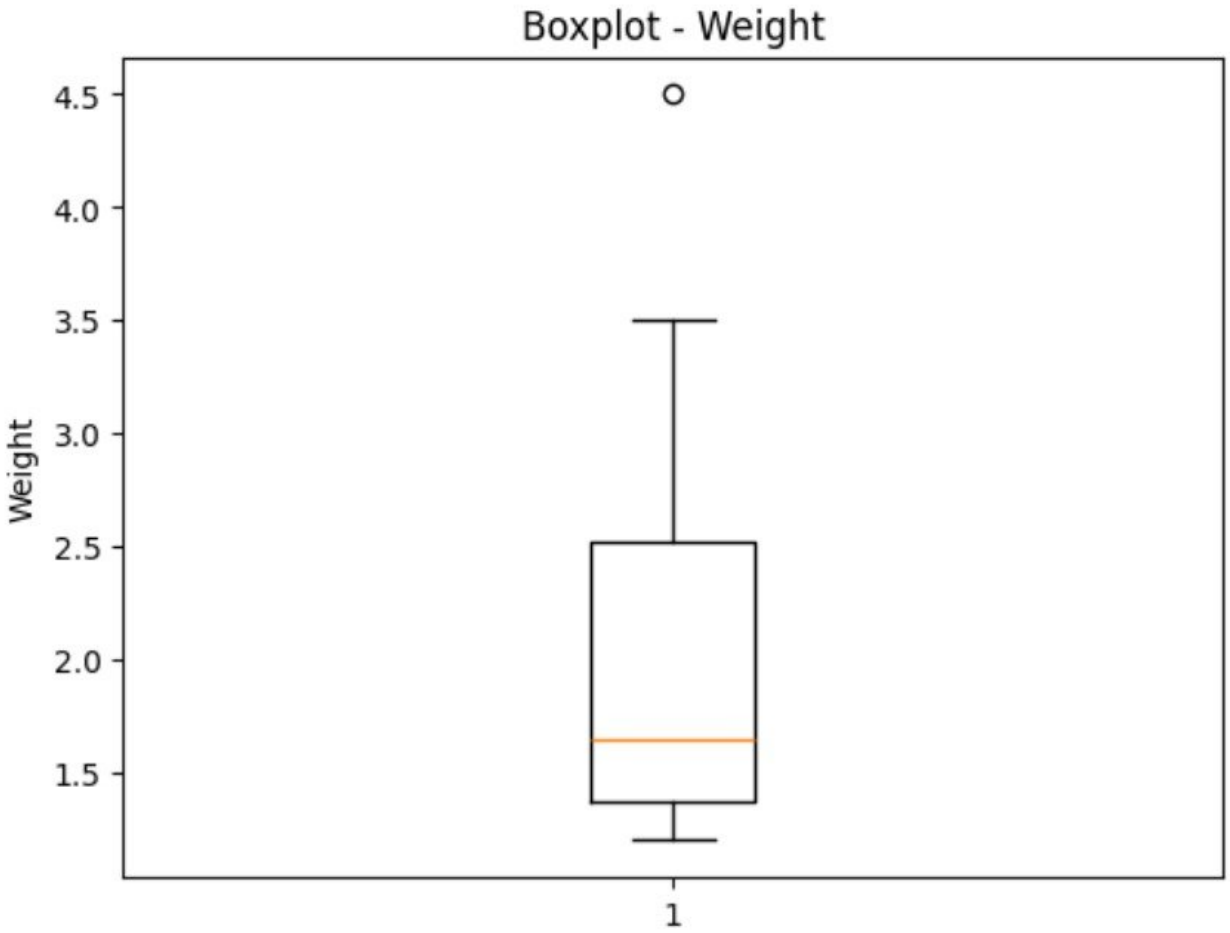
```python
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("laptop.csv")

print("Dataset:")
print(df)
print("\nBoxplot for Screen Size")

plt.boxplot(df["Screen"], notch=True, vert=False, showmeans=True)
plt.xlabel("Screen")
plt.title("Boxplot - Screen Size")
plt.show()
print("\nBoxplot for Weight")

plt.boxplot(df["Weight"])
plt.ylabel("Weight")
plt.title("Boxplot - Weight")
plt.show()


print("\nSeaborn Boxplot for Weight")

sns.boxplot(x=df["Weight"])
plt.title("Seaborn Boxplot - Weight")
plt.show()

print("\nSeaborn Boxplot - Weight vs RAM")
```

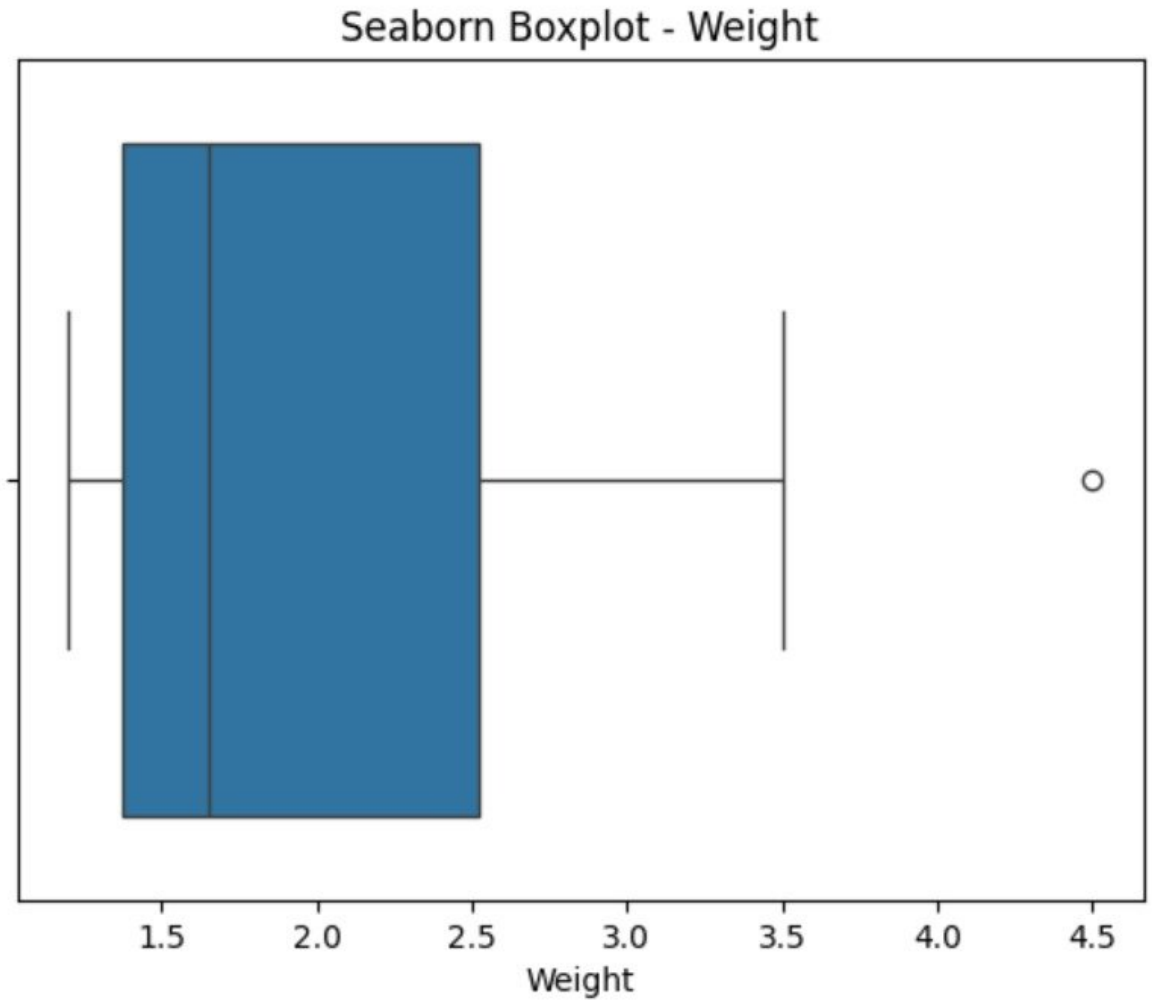
```python
print("\nSeaborn Boxplot - Weight vs RAM")

sns.boxplot(x=df["Weight"], y=df["RAM"])
plt.title("Weight vs RAM")
plt.show()

print("\nIQR Method Calculation")

Q1 = df["Weight"].quantile(0.25)
Q3 = df["Weight"].quantile(0.75)

print("Q1:", Q1)
print("Q3:", Q3)

IQR = Q3 - Q1
print("IQR:", IQR)

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

print("Lower Bound:", lower)
print("Upper Bound:", upper)
outliers = df[(df["Weight"] < lower) | (df["Weight"] > upper)]

print("\nOutliers in Weight:")
print(outliers)
```

```
Dataset:
      Name  Screen  Weight  RAM   Price
0     Dell    15.6     1.8    8   55000
1       HP    14.0     1.5    8   50000
2   Lenovo    15.6     2.2   16   65000
3     Asus    13.3     1.3    8   48000
4     Acer    17.0     3.5   16   70000
5    Apple    13.0     1.2    8   90000
6      MSI    15.6     4.5   32  120000
7  Samsung    14.0     1.4    8   52000

Boxplot for Screen Size
```

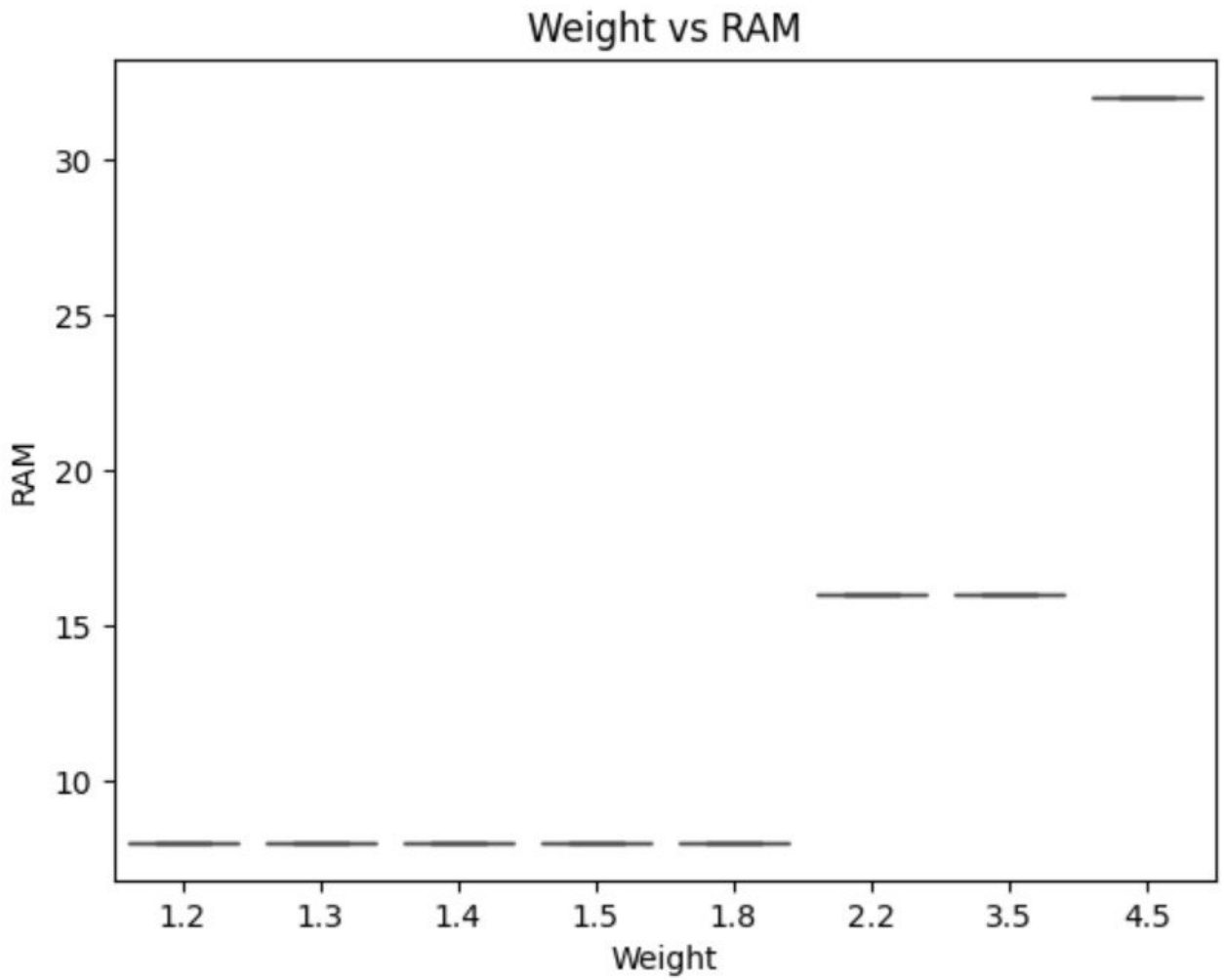


```
Boxplot for Weight
```



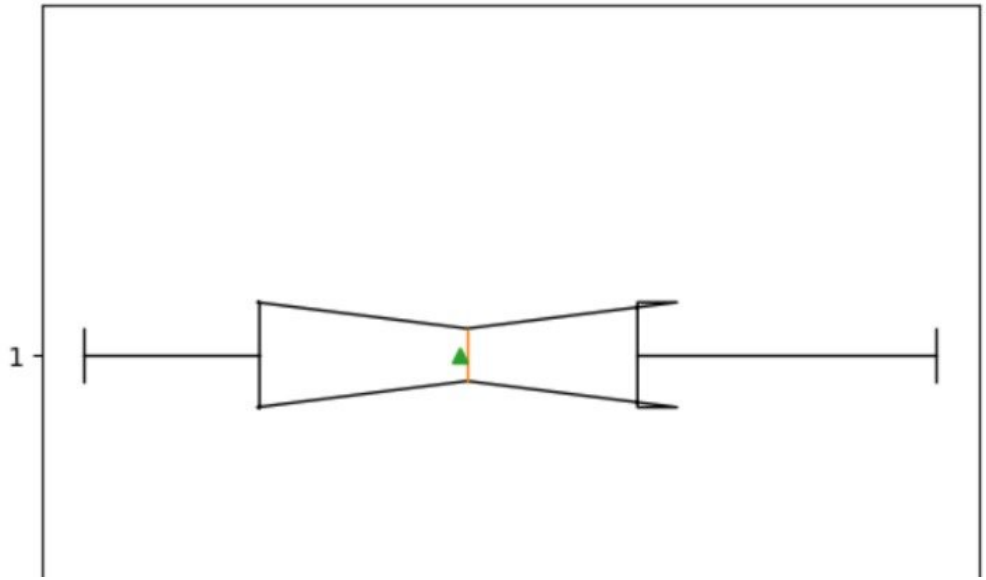

Seaborn Boxplot - Weight
1.5
2.0
2.5
3.0
3.5
4.0
4.5
Weight

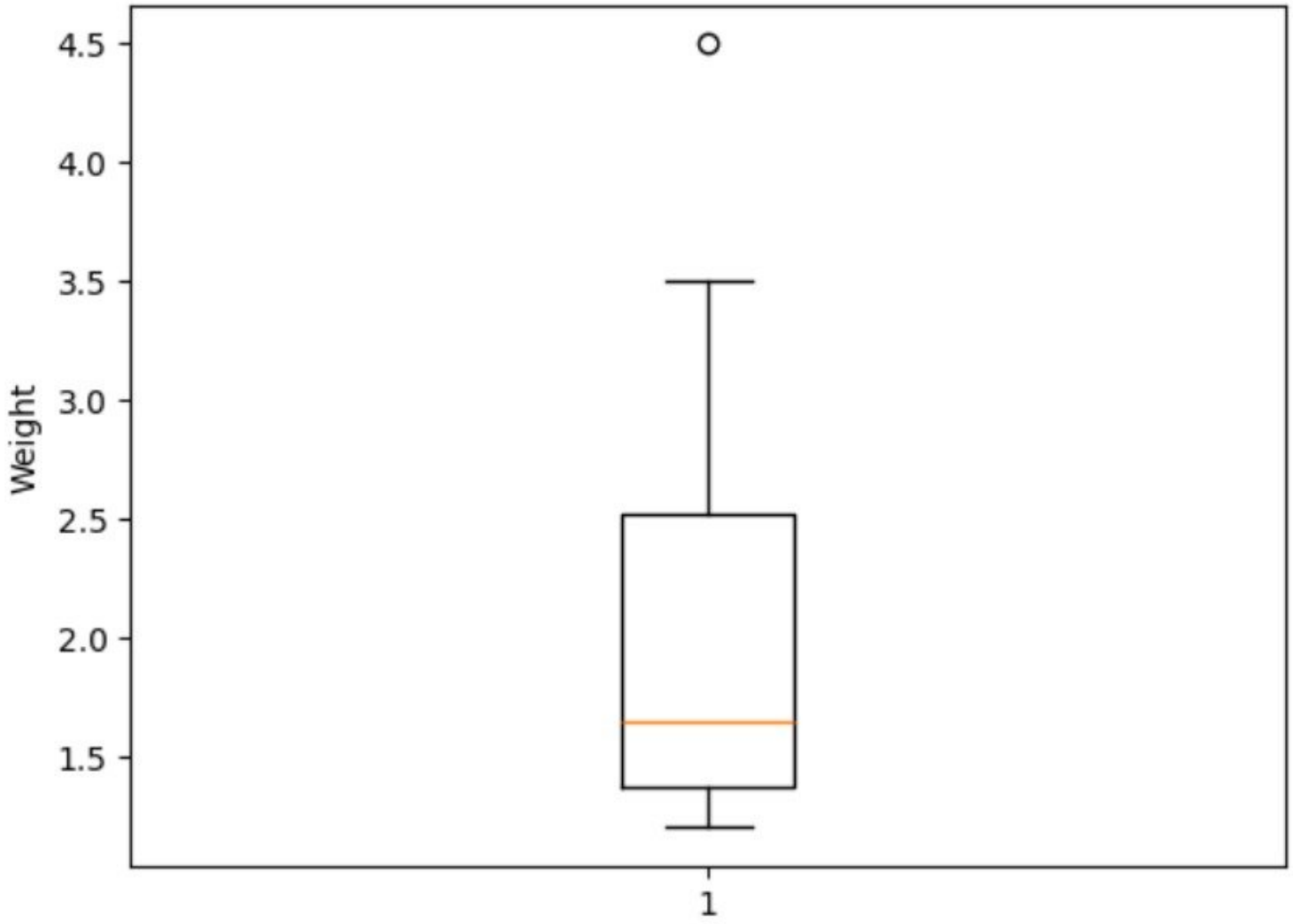
```
Seaborn Boxplot - Weight vs RAM
```



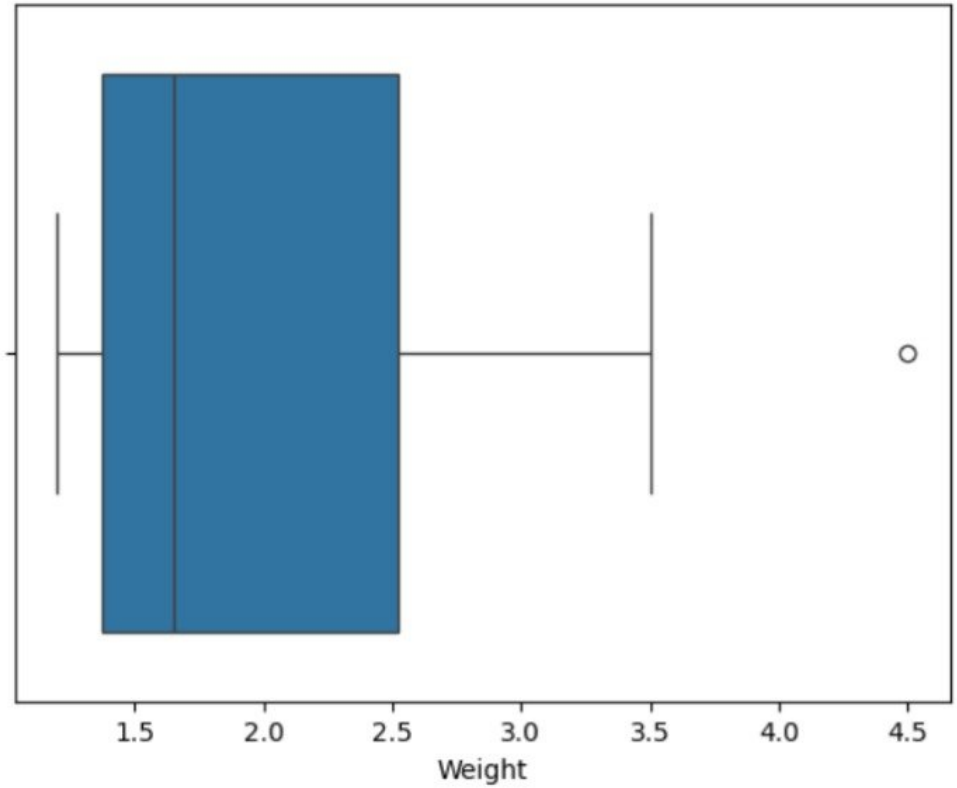
```
IQR Method Calculation
Q1: 1.375
Q3: 2.5250000000000004
IQR: 1.1500000000000004
Lower Bound: -0.35000000000000053
Upper Bound: 4.250000000000001

Outliers in Weight:
  Name  Screen  Weight  RAM   Price
6  MSI    15.6     4.5   32  120000
```

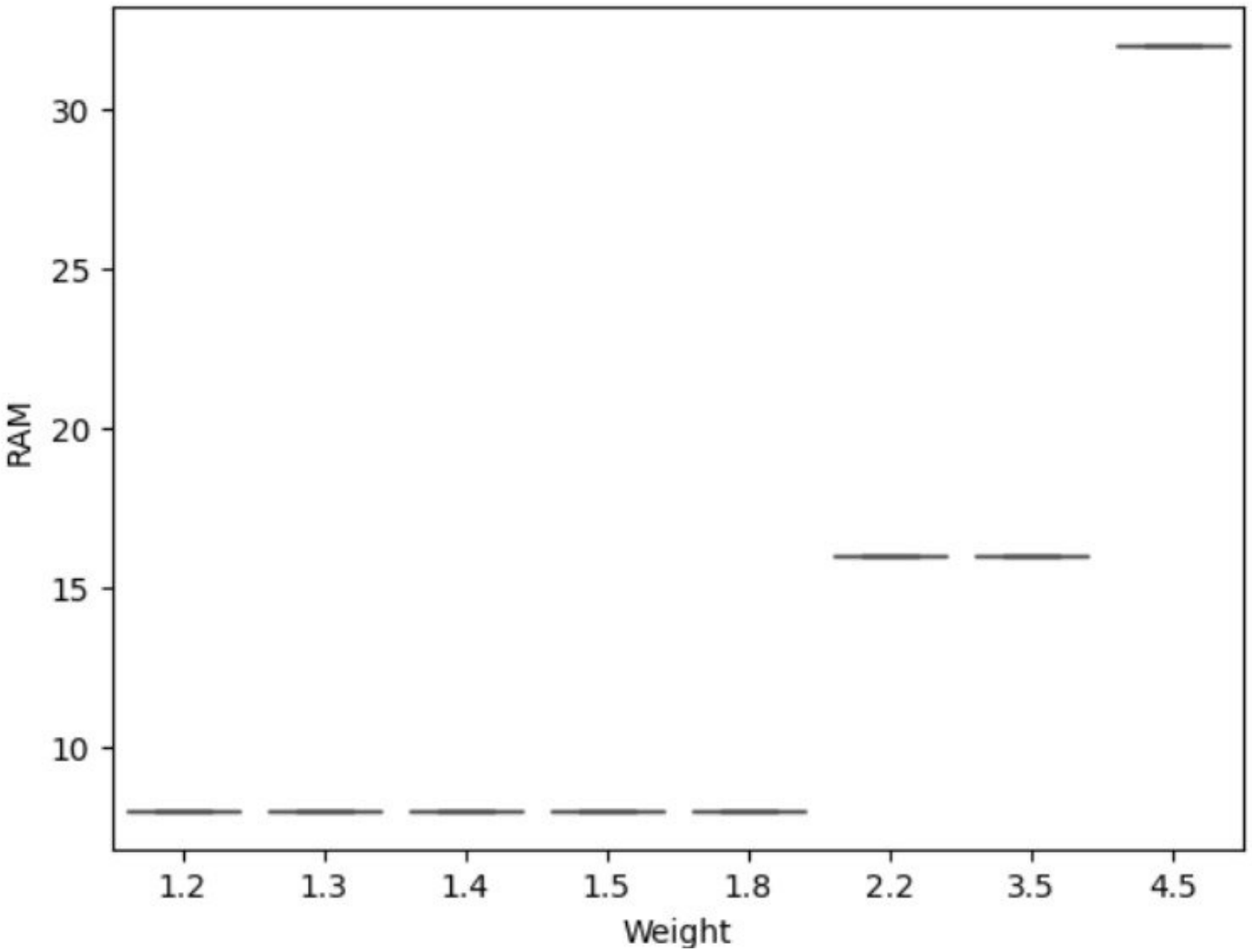
[4]:
```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("laptop.csv")

plt.boxplot(df["Screen"], notch=True, vert=False, showmeans=True)  # notch(curve) around the median
plt.xlabel("Screen")
plt.show()

plt.boxplot(df["Weight"])
plt.ylabel("Weight")
plt.show()
```




4.5
4.0
3.5
3.0
2.5
2.0
1.5
Weight
1

[5]:
```
import seaborn as sns
import matplotlib.pyplot as plt

sns.boxplot(x=df["Weight"])
plt.show()

sns.boxplot(x=df["Weight"], y=df["RAM"])
plt.show()
```




30
25
20
15
10
RAM
1.2
1.3
1.4
1.5
1.8
2.2
3.5
4.5
Weight

[6]:
```
Q1 = df["Weight"].quantile(0.25)
Q3 = df["Weight"].quantile(0.75)

IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

outliers = df[(df["Weight"] < lower) | (df["Weight"] > upper)]

print(outliers)
```

```
  Name  Screen  Weight  RAM   Price
6  MSI    15.6     4.5   32  120000
```